

7(a)	any six from: 1 sodium–potassium pumps (in axon membrane) ; 2 transport, sodium ions / Na^+, out and, potassium ions / K^+, in ; 3 by active transport / uses ATP ; 4 <u>three</u>, sodium ions / Na^+, and <u>two</u>, potassium ions / K^+ ; 5 sets up an electrochemical gradient ; 6 potassium ions / K^+, diffuse out through channels ; 7 membrane more permeable to K^+ / membrane has more K^+ channels ; 8 (so) more, potassium ions / K^+, move out of axon than, sodium ions / Na^+, move in ; 9 <i>ref. to</i> anions / negatively charged ions / negatively charged proteins, inside ; 10 inside negative (relative to outside) ; 11 -60 mV to -70 mV ;	6
------	--	---

Question	Answer	Marks																																								
7(b)(i)	any three from: <i>hypokalaemia</i> ora for normal 1 longer, action potential / depolarisation / repolarisation or larger depolarisation ; 2 longer hyperpolarisation or more negative / smaller, hyperpolarisation ; 3 longer refractory period ; 4 more negative / lower, resting potential ; 5 data quote ; <table><tr><td></td><td></td><td>normal</td><td>hypokalaemia</td></tr><tr><td>mp1</td><td>longer action potential</td><td>2.6 ms</td><td>3.25 ms</td></tr><tr><td>mp1</td><td>longer depolarisation</td><td>1.0 ms</td><td>1.25 ms</td></tr><tr><td>mp1</td><td>longer repolarisation</td><td>1.0 ms</td><td>1.3 ms</td></tr><tr><td>mp1</td><td>larger depolarisation</td><td>110 mV</td><td>140 mV</td></tr><tr><td>mp2</td><td>longer hyperpolarisation</td><td>0.7 ms</td><td>1.0 ms</td></tr><tr><td>mp2</td><td>more negative</td><td>−92 mV</td><td>−112 mV</td></tr><tr><td></td><td>hyperpolarisation</td><td>−22 mV</td><td>−12 mV</td></tr><tr><td>mp2</td><td>smaller hyperpolarisation</td><td></td><td></td></tr><tr><td>mp4</td><td>resting potential</td><td>−70 mV</td><td>−100 mV</td></tr></table>			normal	hypokalaemia	mp1	longer action potential	2.6 ms	3.25 ms	mp1	longer depolarisation	1.0 ms	1.25 ms	mp1	longer repolarisation	1.0 ms	1.3 ms	mp1	larger depolarisation	110 mV	140 mV	mp2	longer hyperpolarisation	0.7 ms	1.0 ms	mp2	more negative	−92 mV	−112 mV		hyperpolarisation	−22 mV	−12 mV	mp2	smaller hyperpolarisation			mp4	resting potential	−70 mV	−100 mV	3
		normal	hypokalaemia																																							
mp1	longer action potential	2.6 ms	3.25 ms																																							
mp1	longer depolarisation	1.0 ms	1.25 ms																																							
mp1	longer repolarisation	1.0 ms	1.3 ms																																							
mp1	larger depolarisation	110 mV	140 mV																																							
mp2	longer hyperpolarisation	0.7 ms	1.0 ms																																							
mp2	more negative	−92 mV	−112 mV																																							
	hyperpolarisation	−22 mV	−12 mV																																							
mp2	smaller hyperpolarisation																																									
mp4	resting potential	−70 mV	−100 mV																																							

Question	Answer	Marks
7(b)(ii)	any two from: 1 harder stimulus / larger stimulus / larger depolarisation, needed, to reach <u>threshold</u> required / to generate an action potential ; 2 fewer, impulses / action potentials, transmitted or lower frequency of impulses ; 3 slower, nervous coordination / reaction times / responses / cognition or weaker, sensation / muscle contraction ; <i>allow named examples</i>	2

Question	Answer	Marks
----------	--------	-------